CivicSaathi CaseLink

Software Requirements Specification (SRS) and Version 1 Implementation Plan

Project category: AI-powered civic grievance management and community issue reporting
Proposed pilot location: Gwalior, Madhya Pradesh, India
Version: 1.0 — Initial prototype
Core principle: One citizen problem, multiple departments, one accountable case.

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1. Problem Statement

Citizens often struggle to identify the correct authority for a civic issue, prepare complaints, navigate different government portals, and determine whether an issue has already been reported. Problems involving multiple departments may be fragmented across separate complaints, making follow-up difficult.

CivicSaathi CaseLink aims to simplify this process through a unified platform that helps citizens report problems, understand the next steps, identify the relevant authority, discover potentially related reports, and track complaint progress.

The system will use AI for image and text analysis, complaint classification, guidance, and potential duplicate detection. It will maintain a unified internal case record while allowing one case to contain multiple department-specific tasks. Official complaint submission and status updates will be treated as verified only when supported by an actual confirmation or an authorized integration.

2. Objectives

1. Make civic grievance reporting easier and more accessible.
2. Help citizens identify the appropriate department and complaint procedure.
3. Use AI to extract useful information from text and images.
4. Identify potentially related reports without automatically merging separate incidents.

5. Maintain a unified case view for issues involving multiple departments.
6. Support complaint preparation, official submission where feasible, and progress tracking.
7. Encourage community participation through issue reporting, upvotes, comments, and updates.
8. Provide a transparent, reliable system that distinguishes verified information from AI-generated suggestions.

3. Users and Stakeholders

User	Responsibilities and needs
Citizen	Report issues, upload evidence, receive guidance, review suggested matches, submit or track complaints.
Community member	View public issues, upvote reports, comment, follow updates, and report inappropriate content.
Case coordinator or moderator	Review flagged content, manage duplicate suggestions, and coordinate internal case records.
Authorized department official	Where access is available, receive assigned tasks, review complaints, and update official progress.
System administrator	Manage users, categories, configuration, access permissions, and system health.
AI service	Analyze submitted content and generate classifications, summaries, guidance, and potential matches.

Important: Department-official access is conditional on authorized participation or an available integration. The prototype must not imply that government departments are connected when they are not.

4. Inputs and Outputs

4.1 Inputs

- Issue description written by the citizen.
- Optional photographs or other supported evidence.
- Issue category, location, and time, where known.

- Optional contact details and relevant complaint information.
- Answers to guided questions for personal or administrative grievances.
- Citizen confirmation or correction of AI-generated suggestions.
- Existing app reports and their locations, timestamps, categories, and statuses.
- Official complaint references, acknowledgements, and status updates, when available.
- Community actions such as upvotes, comments, follows, and moderation reports.
- Trusted news sources and publication metadata, if the nearby-news module is enabled.

4.2 Outputs

- Suggested issue category and structured complaint summary.
- Recommended authority or official complaint channel.
- Step-by-step guidance and a draft complaint.
- A unified internal CaseLink case ID.
- Potentially related reports, with similarity explanations.
- Department-specific tasks linked to a single internal case.
- Official filing confirmation and reference number, only when genuinely received.
- Case timeline, notifications, and status history.
- Public issue feed, community engagement counts, and verified-source news summaries where supported.
- Error messages and alternative instructions when an operation fails.

5. System Workflow

Step 1 — Report an issue: The citizen enters a description, optionally uploads a photograph, and provides a location or other relevant details.

Step 2 — AI-assisted analysis: The system extracts relevant details, suggests an issue category, identifies missing information, and generates a concise summary. The citizen can correct the results.

Step 3 — Identify the responsible authority: The system recommends a relevant department or official complaint channel using configured rules and verified reference information.

Step 4 — Check for related reports: The platform searches existing reports for possible matches based on location, category, text similarity, and time. It displays potential matches without automatically merging them.

Step 5 — Create the internal case: After the citizen confirms the report, the system creates a CaseLink case with an internal case ID, evidence, timeline, and status.

Step 6 — Coordinate tasks: If multiple departments may be involved, the case can contain separate linked tasks for each department. Each task maintains its own status and evidence.

Step 7 — Prepare and submit the complaint: The system prepares a complaint draft and guides the citizen through the relevant official process. Authorized integrations may submit it directly where supported. Otherwise, the citizen completes the official submission through the supported channel.

Step 8 — Record confirmation: The system records the official reference number and acknowledgement only after receiving or obtaining confirmation. Otherwise, the case remains marked as awaiting citizen submission or confirmation.

Step 9 — Track and update: Citizens view the internal case timeline and any confirmed official status updates. Community updates are displayed separately from official responses.

Step 10 — Resolve or close: A case can be marked resolved based on defined evidence or an authorized status update. Citizens may provide feedback or flag an incorrect resolution.

6. Functional Requirements

ID	Requirement	Version 1 priority

FR-01	Users can register, sign in, and access their own cases securely.	Must
FR-02	Citizens can create an issue report using text, location, category, and optional images.	Must
FR-03	AI can analyze supported images and descriptions to suggest issue details and categories.	Must
FR-04	The system can generate a structured complaint summary and actionable guidance.	Must
FR-05	The system can recommend an appropriate authority or complaint channel from configured reference data.	Must
FR-06	The system generates a unique internal CaseLink ID and maintains a case timeline.	Must
FR-07	The system identifies potential duplicate or related reports and allows human review.	Must
FR-08	A case can contain multiple linked department tasks without losing the original case history.	Must
FR-09	The system can prepare complaint data and support a guided official filing workflow.	Must
FR-10	Official ticket numbers and statuses are recorded only when supported by evidence or an authorized integration.	Must
FR-11	Users can view public issues, upvote, comment, follow, and report inappropriate content, subject to privacy and moderation rules.	Should
FR-12	Users can view a daily or nearby news feed with source and publication date, where reliable data is available.	Should
FR-13	The system provides an emergency-help interface with appropriate instructions and verified contact options.	Must
FR-14	Administrators can manage issue categories, configured authority information, and flagged content.	Must

FR-15	The system records key status changes and handles failed submissions without falsely marking them successful.	Must

Scope clarification: “Must” means essential for the proposed prototype. If an external government portal does not offer a usable, authorized integration, FR-09 will be implemented through complaint preparation and guided submission rather than an unverified claim of automatic filing.

7. AI Responsibilities

AI will act as an assistant, not as the final authority.

1. **Image understanding:** Identify visible features relevant to a reported issue, such as a pothole, accumulated waste, damaged infrastructure, or a water leak. It must acknowledge uncertainty and avoid claiming facts that cannot be established from the image.
2. **Classification:** Suggest the appropriate issue category and extract relevant details from text and images.
3. **Information gathering:** Identify missing details and ask follow-up questions before preparing an incomplete complaint.
4. **Complaint assistance:** Generate readable complaint drafts and explain possible next steps.
5. **Authority recommendation:** Recommend a department or channel using configured rules and trusted reference information.
6. **Potential duplicate detection:** Rank similar reports and explain the factors contributing to a match.
7. **Case coordination:** Suggest which departments may need separate tasks and produce a concise case summary.
8. **News assistance:** Summarize available articles while retaining source links and publication dates.

AI must not invent official policies, contact numbers, ticket IDs, complaint outcomes, news events, or government integrations. Low-confidence results should be labelled as uncertain.

8. Human Responsibilities

- Citizens verify the report, location, evidence, and complaint details before submission.
- Citizens approve consequential actions, including official complaint submission where required.
- Citizens review suggested duplicate matches and can indicate whether two reports concern the same incident.
- Moderators handle disputed matches, harmful content, spam, and reports requiring human review.
- Authorized officials make official decisions and provide authoritative status updates.
- Administrators verify and maintain department directories, official links, and configured filing procedures.
- Human operators handle cases requiring judgement beyond the AI's reliable capabilities.

AI-generated recommendations must not independently determine guilt, legal liability, eligibility for benefits, or the validity of a harassment allegation.

9. Non-Functional Requirements and Constraints

9.1 Security and privacy

- Users must not be able to access another citizen's private case or personal documents without authorization.
- Sensitive complaints, including harassment, must be private by default.
- Public community posts must not expose personal contact details, private evidence, or sensitive complaint narratives.
- Access controls, secure password handling, input validation, and protected data transmission must be implemented.
- Collect only information necessary for the stated purpose and define retention and deletion procedures.

9.2 Reliability and transparency

- Internal case IDs must be distinguishable from official government reference numbers.

- The system must distinguish AI suggestions, citizen-submitted information, community reports, and verified official updates.
- Every consequential status change should have a timestamp and recorded source.
- Failures must not silently discard a report or mark an unsuccessful operation as completed.

9.3 Usability

- The reporting flow should use simple language and clear steps.
- Citizens should be able to correct AI suggestions.
- Error messages should explain the problem and offer a reasonable next action.
- The interface should work on common mobile and desktop screen sizes.

9.4 External constraints

- Government portal integration depends on available APIs, permission, authentication, and the portal's terms.
- CAPTCHA, OTP, authentication, and other access controls must not be bypassed.
- Image-analysis quality depends on image clarity, lighting, and model capability.
- Location-based matching depends on the accuracy of submitted locations.
- News freshness depends on the availability and reliability of news sources.
- The prototype's pilot geography and supported issue categories must be explicitly configured rather than assumed to cover every region.

10. Failure Handling

Failure scenario	Expected system response
Image is unclear or unsupported	Explain the limitation and allow a text-only report or another upload.

AI service is unavailable	Preserve the submitted information and offer manual category selection and complaint preparation.
Required information is missing	Ask for the missing details or clearly mark them as unknown.
No suitable authority can be identified	Present a review-needed state and available verified general guidance instead of inventing a destination.
Similar reports are found	Show possible matches for review; do not merge automatically.
Official portal or integration is unavailable	Preserve the draft, explain that submission has not been confirmed, and provide the official channel when verified.
Filing is interrupted	Retain the draft where safe and allow the citizen to retry or resume.
Official acknowledgement is missing	Keep the status as unconfirmed; do not generate a ticket number.
Location is unavailable or inaccurate	Allow manual location entry or an approximate location, clearly labelled.
News data is unavailable or stale	Show a data-unavailable message or the last verified update time; do not fabricate current news.
Emergency contact or dispatch integration is unavailable	Display verified emergency guidance and clearly state whether any call or alert was actually initiated.
Malicious or private community content is reported	Restrict exposure where appropriate and queue it for moderation.

11. Acceptance Criteria

The Version 1 prototype will be considered acceptable when the following tests pass.

1. **Account and privacy:** A user can create an account, sign in, create a case, and view it again. An unauthorized user cannot access the private case.
2. **Issue reporting:** A citizen can submit a text-based report with a category and location, with an optional image.
3. **AI assistance:** For a supported test image and description, the system returns a structured suggestion or a clear uncertainty/error response. The citizen can edit the result.
4. **Complaint preparation:** The system generates a draft containing the supplied facts without inventing missing details.
5. **Authority guidance:** A test issue maps to a configured authority or is explicitly flagged for review when no reliable mapping exists.
6. **Case tracking:** Every accepted report receives a unique internal CaseLink ID and a persistent timeline.
7. **Linked departments:** A test case can contain two linked department tasks, each with its own status, while remaining part of one case.
8. **Duplicate detection:** A similar report is surfaced as a possible match, and the citizen or authorized moderator can review it without automatic merging.
9. **Filing integrity:** The system records an official reference only after actual confirmation. A failed or unconfirmed submission is never displayed as successfully filed.
10. **Community features:** Authorized users can view public reports, upvote, comment, and report inappropriate content without gaining access to private cases.
11. **Emergency safety:** The emergency interface displays verified instructions and does not claim that emergency services have been contacted unless this is confirmed.
12. **Failure recovery:** If the AI or an external filing service is unavailable, the report is not silently lost and the interface communicates the failure.

13. **News provenance:** Any displayed news item includes a source and publication date or timestamp. If reliable current data is unavailable, the system states that clearly.
14. **Test evidence:** Core workflow tests, access-control tests, and failure-path tests are run and documented before the demonstration.

For the initial pilot, the team should demonstrate the complete internal reporting-to-tracking journey end to end. External government filing should be demonstrated only if the integration has been genuinely implemented and tested.

12. Version 1 Exclusions

The following features are outside the initial prototype's guaranteed scope:

- Universal integration with every government department or portal.
- Guaranteed automatic filing on websites that do not provide an authorized, usable integration.
- Automatic resolution of civic issues or control over government response times.
- Guaranteed emergency-service dispatch or live emergency response coordination.
- Automatic merging of citizen reports without review.
- Automated legal judgements, eligibility decisions, or accusations against individuals.
- Guaranteed image-based identification of every type of civic problem.
- Nationwide deployment and support for every language and jurisdiction.
- Guaranteed real-time news coverage without a dependable data source.
- Predictive claims about when an authority will resolve an issue.
- Facial recognition or biometric identity verification.
- Claims of official government endorsement or partnership without authorization.

These exclusions keep the prototype achievable while leaving room for later expansion.

13. Implementation Plan

Phase 1 — Requirements and foundation

- Finalize the pilot geography, supported categories, user roles, and workflows.
- Review official complaint channels and record their supported submission methods.
- Set up the application architecture, database, authentication, and basic interface.
- Define the internal case, department task, evidence, and status models.

Deliverable: Running application foundation and documented data model.

Phase 2 — Core case management

- Implement issue submission, case IDs, private case views, and case timelines.
- Add configurable authority recommendations.
- Implement linked department tasks and status tracking.

Deliverable: End-to-end internal grievance workflow.

Phase 3 — AI assistance

- Add image and text analysis, category suggestions, missing-information prompts, and complaint drafting.
- Add potential duplicate detection and human review.
- Provide manual fallbacks for AI failures.

Deliverable: AI-assisted reporting and case coordination.

Phase 4 — Filing and community

- Implement official-channel guidance and the supported complaint-submission workflow.
- Add acknowledgement capture and clearly separated internal and official statuses.
- Implement public reports, upvotes, comments, follows, and moderation.

- Add the emergency-help interface and source-linked news module where reliable data is available.

Deliverable: Citizen-facing MVP with honest integration status.

Phase 5 — Testing and demonstration

- Test the full workflow using representative civic issue scenarios.
- Test duplicate suggestions, private cases, failed AI requests, and failed external submissions.
- Check for invented ticket numbers, unsupported claims, and incorrect success messages.
- Fix critical defects and document any incomplete integrations.

Deliverable: Tested prototype, acceptance-test evidence, known-limitations list, and demonstration script.

14. Suggested Demonstration Scenario

A citizen uploads a photograph of a pothole and reports its location.

1. CivicSaathi suggests the category and generates a concise description.
2. The citizen reviews and confirms the details.
3. The system identifies a potentially related report nearby.
4. The citizen reviews the match, and the platform creates or updates the appropriate internal case without automatically merging incidents.
5. CaseLink recommends the configured official complaint channel and prepares the complaint.
6. The citizen completes the official filing process, or an authorized integration submits it.
7. The system records the official reference only when confirmation is received.
8. The citizen follows the internal case timeline and any confirmed official status updates.

This scenario demonstrates the central value proposition: **a unified, transparent citizen experience across reporting, coordination, and tracking,**

without pretending that AI replaces human judgement or government authority.

15. Success Measures for the Pilot

The team will measure:

- Percentage of test reports successfully created and retrievable.
- Percentage of supported test cases receiving a useful category and authority recommendation.
- Accuracy of potential duplicate suggestions on a labelled test set.
- Percentage of official submission attempts correctly classified as confirmed, failed, or unconfirmed.
- Number of critical privacy or access-control defects found during testing.
- Completion rate of the reporting-to-tracking demonstration.
- Percentage of displayed news items with traceable sources and publication metadata.

Targets should be set after the team establishes a baseline on a small, representative test dataset. The prototype should report measured results rather than claiming untested accuracy.

Final implementation principle: Build a reliable case-management foundation first, add AI where it measurably improves the workflow, and integrate external government systems only where the necessary access and confirmation mechanisms genuinely exist.